CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)

CSPIT — Department of Computer Engineering · ITUE203 Web Development Frameworks

Student Name	Dhruv Bhoi	**Student ID**	**D26CE154**
Academic Year	2026–27 (ODD)	**Semester**	3rd Semester, B.Tech CE

Practical 4: JavaScript DOM Manipulation, Event Handling, and UI Interactivity

Practical Aim & Tools

Aim: Implement UI interactivity, DOM manipulation, event handling, dark/light theme switcher with localStorage, accordion FAQ, modal popups, banner notifications, and carousel slider.

Tools & Technologies: Vanilla ES6+ JavaScript, DOM APIs, Event Delegation, HTML5 LocalStorage

1. DOM Interactivity Modules

StudentHub features lightweight, dependency-free Vanilla ES6+ JavaScript (`assets/js/main.js`) powering five key UI features:
1. Theme Switcher: Reads and writes user preference (`light` or `dark`) to `localStorage`. An inline IIFE prevents theme flash on page load.
2. Collapsible Accordion: Implemented in `faq.php` with keyboard accessibility and ARIA expanded state management.
3. Accessible Modal Dialog: Accessible trap focus dialog for Quick Tours and Registration previews.
4. Sliding Highlights Carousel: Automated and keyboard-accessible slide transitions on the homepage.
5. Dismissible Status Banner: Persistent notification banner with smooth dismissal animations.

2. Implementation Code: Theme Toggle & Accordion

Listing: main.js — Theme Switcher and Accessible Accordion

```
// Theme Toggle with LocalStorage Persistence
const themeToggle = document.getElementById('themeToggle');
themeToggle?.addEventListener('click', () => {
  const current = document.documentElement.getAttribute('data-theme') || 'light';
  const target = current === 'dark' ? 'light' : 'dark';
  document.documentElement.setAttribute('data-theme', target);
  localStorage.setItem('sh_theme', target);
  themeToggle.textContent = target === 'dark' ? '🌞' : '🌜';
});

// Accessible FAQ Accordion
```

```
document.querySelectorAll('.accordion-header').forEach(btn => {
  btn.addEventListener('click', () => {
    const expanded = btn.getAttribute('aria-expanded') === 'true';
    btn.setAttribute('aria-expanded', !expanded);
    const body = btn.nextElementSibling;
    body.hidden = expanded;
  });
});
```

3. Verification Screenshot: Interactive FAQ Accordion

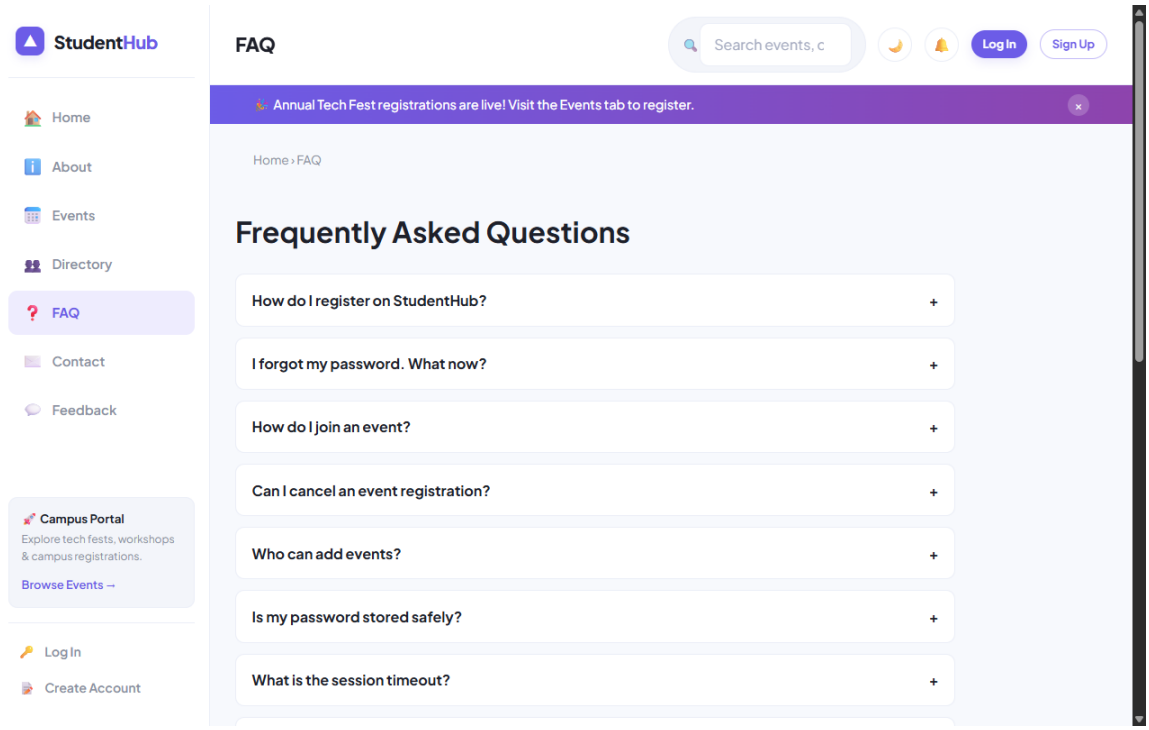


Figure: Interactive FAQ Accordion with Search and Dark/Light Mode Support

4. Key Questions & Analysis

Q: How are DOM elements selected and modified?

Ans: Elements are selected using modern `document.querySelector()` and `document.querySelectorAll()`. Modifications use secure DOM APIs like `element.textContent`, `element.setAttribute()`, and `element.classList.toggle()` to prevent cross-site scripting (XSS) risks.

Q: Is localStorage used for remembering user theme preferences?

Ans: Yes. When the theme toggle button is clicked, `localStorage.setItem('sh_theme', target)` persists the state. An immediate synchronous inline script in `<head>` executes before the DOM renders to prevent visual flickering.

Practical 4 Conclusion & Learning Outcomes

Implemented rich client-side interactivity using Vanilla ES6+ JavaScript. Features include persistent dark/light theme switching via localStorage, accessible FAQ accordion panels, modal dialog popups with focus traps, and animated highlight sliders.